CLASS 8 · JEE ADVANCED · TOP-10 QUESTION IMPACT REPORT

Pragya G8 (EXI) Achievers

Part Test-1 · Paper 2 · which questions cost the top 10 the most

Exam date 09 Oct 2026
Campus HYD/Nallakunta DD Colony
Students analysed top 10 by program rank

In total, the top 10 lost **≈ 633 marks** – **≈ 128** from Maths, **≈ 302** from Physics and **≈ 203** from Chemistry.

Maths 20% | Physics 48% | Chemistry 32%

Marks lost = marks not earned on wrong or skipped questions plus negative marks. Approximate (≈) because multiple-correct questions give partial marks.

17 of 46
high-impact questions – lost by 5+ of the top 10

48%
of all lost marks came from Physics

Q27 10/10
lost by every student – plus 4 more questions at 10/10

96 / 180
top-10 average score (range 82–117)

1 Where the damage happened

Marks the top 10 gave away in each subject, split into marks lost on **wrong answers** (the question's marks plus the negative mark) and marks left on **skipped** questions. Advanced figures are approximate (≈) – see note at the end.

SUBJECT	SPLIT (wrong / skipped)	MARKS LOST	SHARE	ACCURACY*	HIGH-IMPACT
Maths	≈ 122 on 32 wrong answers (incl. 14 negative) · 6 on 2 skipped	≈ 128 (≈ 13 per student)	20%	77%	4 of 14 Qs
Physics	≈ 272 on 71 wrong answers (incl. 25 negative) · 30 on 9 skipped	≈ 302 (≈ 30 per student)	48%	53%	9 of 16 Qs
Chemistry	≈ 182 on 45 wrong answers (incl. 28 negative) · 21 on 7 skipped	≈ 203 (≈ 20 per student)	32%	71%	4 of 16 Qs

* Accuracy = right answers ÷ questions attempted.

BY QUESTION TYPE	MATHS	PHYSICS	CHEMISTRY
Single correct (marking +3 / −1)	Q1–4 **15%** wrong · **0%** skipped	Q18–21 **22%** wrong · **8%** skipped	Q35–38 **20%** wrong · **8%** skipped
Multiple correct (marking +4 / −2)	Q5–7 **13%** wrong · **0%** skipped	Q22–24 **27%** wrong · **7%** skipped	Q39–41 **33%** wrong · **0%** skipped
Integer (marking +4 / 0)	Q8–13 **16%** wrong · **0%** skipped	Q25–30 **52%** wrong · **2%** skipped	Q42–47 **15%** wrong · **0%** skipped
Paragraph (marking +3 / 0)	Q14–17 **70%** wrong · **10%** skipped	Q31–34 **70%** wrong · **8%** skipped	Q48–51 **60%** wrong · **13%** skipped

Key takeaways

1. **Physics** did the most damage: **48%** of all marks lost by the top 10 (≈ 302 of ≈ 633). 9 of its 16 questions were lost by 5 or more students.
2. **Q27, Q31, Q32, Q48, Q51** were lost by all 10 students, and **1 more** by 9 of 10 (Q16).
3. Biggest chapter-level loss: **Chemistry · Mole Concept & Basics** – ≈ 121 marks across 9 questions (3 high-impact).
4. **Paragraph** questions were the weakest type in the paper: **77%** of answers wrong or blank, against 36% across all questions.
5. Negative marking alone cost the top 10 **67 marks** (about 7 per student).
6. Strong ground: all 10 answered **11 questions** correctly (Q1, Q2, Q3, Q7, Q8, Q9, Q11, Q22, Q37, Q43, Q45).

Answered wrong by 5+ students

Students attempted these and still got them wrong – points to a misconception or a trap worth re-teaching.

Q27 10 wrong | **Q31** 10 wrong | **Q32** 10 wrong | **Q48** 10 wrong
Q16 9 wrong | **Q4** 6 wrong | **Q10** 6 wrong | **Q30** 6 wrong
Q39 6 wrong | **Q51** 6 wrong | **Q15** 5 wrong | **Q28** 5 wrong

Left blank by 5+ students

Most students did not attempt these – points to an unfamiliar topic, low confidence, or running out of time.

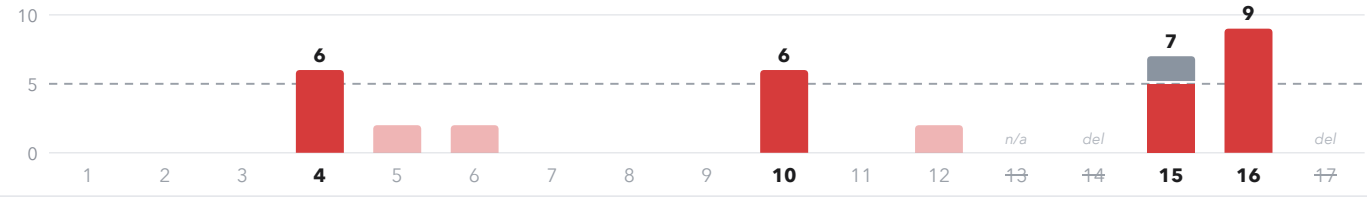
None

2 Question-by-question impact

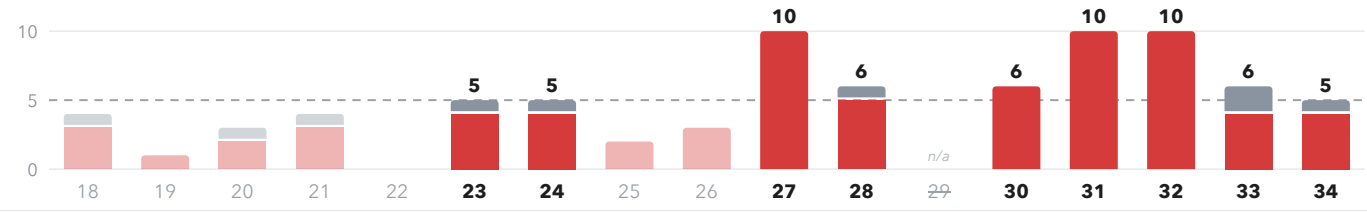
How many of the top 10 lost each question. Questions at or above the dashed line (lost by 5+) are **high-impact** and shown in full colour with their count.

Wrong answer · Skipped · ---- High-impact line (5 of 10) · Below the line (faded) · "del" = deleted / bonus question

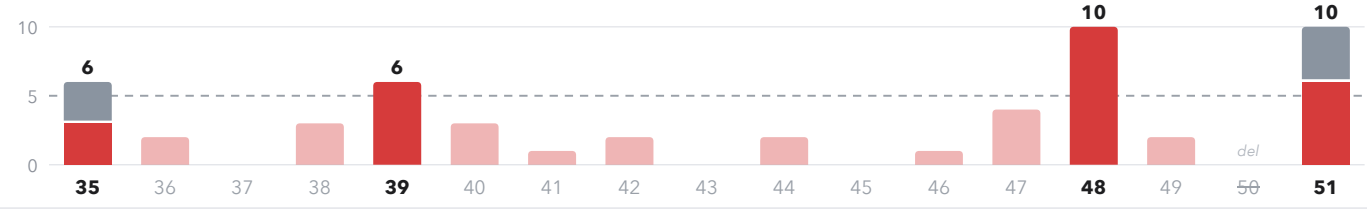
Maths — **4** of 14 questions high-impact · ≈ **128** marks lost by the top 10



Physics — **9** of 16 questions high-impact · ≈ **302** marks lost by the top 10



Chemistry — **4** of 16 questions high-impact · ≈ **203** marks lost by the top 10



3 Chapter hot-spots

Marks lost by the top 10, chapter by chapter. Subjects are ordered from most marks lost to least; within each subject, re-teach from the top.

Marks lost adds up what all 10 students gave away on a chapter's questions: the question's marks plus its negative mark for each wrong answer (single correct 3 + 1 = 4, multiple correct 4 + 2 = 6, integer 4, paragraph 3), and the question's marks for each skipped one. Example: **Chemistry · Mole Concept & Basics** has 9 questions, and the top 10 lost ≈ **121** marks on them – about **12 per student**. **Students per Q** = on average, how many of the 10 lost each question in that chapter (here 3.4 of 10).

CHAPTER	QS	HIGH-IMPACT QUESTIONS	MARKS LOST BY TOP 10	PER STUDENT	STUDENTS PER Q
1. Physics ≈ 302 marks lost · **48%** of all lost marks · **9** high-impact questions					
Laws of Motion	4	Q23, Q24, Q27, Q28	≈ 120	≈ 12 marks	6.5 of 10
Vectors	5	Q31, Q32	≈ 95	≈ 10 marks	5.8 of 10
Kinematics	7	Q30, Q33, Q34	≈ 87	≈ 9 marks	3.6 of 10
2. Chemistry ≈ 203 marks lost · **32%** of all lost marks · **4** high-impact questions					
Mole Concept & Basics	9	Q35, Q39, Q48	≈ 121	≈ 12 marks	3.4 of 10
Atomic Structure	4	Q51	≈ 64	≈ 6 marks	4.2 of 10
Chemical Bonding	1	None reached 5 · most missed Q38 (3/10)	≈ 12	≈ 1 marks	3.0 of 10
Periodic Classification	2	None reached 5 · most missed Q41 (1/10)	≈ 6	≈ 1 marks	0.5 of 10
3. Maths ≈ 128 marks lost · **20%** of all lost marks · **4** high-impact questions					
Logarithms	4	Q4, Q16	≈ 51	≈ 5 marks	3.8 of 10
Quadratic Equations	6	Q10	≈ 48	≈ 5 marks	1.7 of 10
Sequences & Series	4	Q15	≈ 29	≈ 3 marks	2.2 of 10

4 High-impact questions at a glance

Every question lost by 5 or more of the top 10, one table per subject, worst first. The cropped question for each is in section 7.

Marks lost = marks the top 10 together gave away on that question: the question's marks plus its negative mark for each wrong answer (single correct 3 + 1 = 4, multiple correct 4 + 2 = 6, integer 4, paragraph 3), and the question's marks for each skipped one. Example: **Q27** – 10 wrong × 4 = 40 marks. **Lost by** = how many of the 10 got it wrong or skipped it.

Critical lost by 9–10 **High** lost by 7–8 **Moderate** lost by 5–6

Maths

4 of 14 questions high-impact · ≈ **96** marks lost on them by the top 10

Q	CHAPTER – WHAT IT TESTS	TYPE	LOST BY	WRONG	SKIPPED	MARKS LOST (TOP 10)
Q16	**Logarithms** – Log equations – A + B + C	Paragraph · +3 / 0	9/10	9	0	≈ 27
Q15	**Sequences & Series** – AP: ratio of sums of 14 terms	Paragraph · +3 / 0	7/10	5	2	≈ 21
Q4	**Logarithms** – Logs in AP – solve for x	Single · +3 / −1	6/10	6	0	≈ 24
Q10	**Quadratic Equations** – Roots of two linked quadratics	Integer · +4 / 0	6/10	6	0	≈ 24

Physics

9 of 16 questions high-impact · ≈ **237** marks lost on them by the top 10

Q	CHAPTER – WHAT IT TESTS	TYPE	LOST BY	WRONG	SKIPPED	MARKS LOST (TOP 10)
Q27	**Laws of Motion** – Normal reaction between pushed blocks	Integer · +4 / 0	10/10	10	0	≈ 40
Q31	**Vectors** – Carrom passage – displacement centre → edge	Paragraph · +3 / 0	10/10	10	0	≈ 30
Q32	**Vectors** – Carrom passage – displacement edge → hole	Paragraph · +3 / 0	10/10	10	0	≈ 30
Q30	**Kinematics** – a from v(t) at a given position	Integer · +4 / 0	6/10	6	0	≈ 24
Q28	**Laws of Motion** – Spring-connected blocks – acceleration	Integer · +4 / 0	6/10	5	1	≈ 24
Q33	**Kinematics** – Train passage – retardation to avoid collision	Paragraph · +3 / 0	6/10	4	2	≈ 18
Q23	**Laws of Motion** – Block on block, no friction – motion of m	Multiple · +4 / −2	5/10	4	1	≈ 28
Q24	**Laws of Motion** – Tensions in two inclined strings	Multiple · +4 / −2	5/10	4	1	≈ 28
Q34	**Kinematics** – Train passage – time of retardation	Paragraph · +3 / 0	5/10	4	1	≈ 15

Chemistry

4 of 16 questions high-impact · ≈ **117** marks lost on them by the top 10

Q	CHAPTER – WHAT IT TESTS	TYPE	LOST BY	WRONG	SKIPPED	MARKS LOST (TOP 10)
Q48	**Mole Concept & Basics** – Empirical formula from % composition	Paragraph · +3 / 0	10/10	10	0	≈ 30
Q51	**Atomic Structure** – Hydrogen-like ion – find Z from photon energies	Paragraph · +3 / 0	10/10	6	4	≈ 30
Q39	**Mole Concept & Basics** – Meaning of 1 gram-atom of nitrogen	Multiple · +4 / −2	6/10	6	0	≈ 36
Q35	**Mole Concept & Basics** – Limiting reagent over two-step reaction	Single · +3 / −1	6/10	3	3	≈ 21

5 Who lost which high-impact question

Each row is a student (campus order); each column a high-impact question. **Prog. rank** = rank in the program, **AIR** = All-India rank, **Total** = marks out of 180, **Avg** = average of the 6 weekly Advanced tests (WTA), out of 180. ✗ = wrong, – = skipped, ✓ = correct.

✗ Wrong – Skipped ✓ Correct

	Rank & score				Maths				Physics								Chemistry				
STUDENT	PROG. RANK	AIR	TOTAL	AVG /180	4	10	15	16	23	24	27	28	30	31	32	33	34	35	39	48	51	Lost
Ruperao Shritan	14	38	117	122.5	✓	✓	✗	✗	✗	✓	✗	✗	✓	✗	✗	✓	✓	✗	✓	✗	–	10
Kolukula H. S. N. Hariharan	17	44	116	132.7	✗	✓	✓	✓	✓	✓	✗	–	✗	✗	✗	✓	✓	✓	✓	✗	✗	8
Kavuru Sahithi	27	89	104	127.2	✓	✗	✓	✗	✓	✓	✗	✓	✓	✗	✗	✗	✗	✓	✓	✗	✗	9
Fareeha Shaik	29	97	102	90.8*	✓	✓	✓	✗	✓	✓	✗	✓	✗	✗	✗	✗	✗	✓	✓	✗	✗	9
Karthikeya Silver	38	114	95	109.5	✗	✓	✗	✗	✓	✓	✗	✗	✓	✗	✗	✗	✗	–	✗	✗	✗	13
Sriyan Brahmakanti	52	214	86	93.1	✗	✗	✗	✗	✗	✗	✗	✓	✗	✗	✗	✗	✓	✗	✗	✗	–	15
Chitluri Nayan Swaroop	54	200	85	68.9	✓	✗	✗	✗	✓	✗	✗	✗	✗	✗	✗	–	–	✓	✗	✗	✗	14
G Umesh Babu	58	242	84	99.3	✗	✗	✗	✗	✗	✗	✗	✓	✗	✗	✗	✓	✓	✗	✗	✗	–	14
Nakka Mahidhar	59	209	84	98.9	✗	✗	–	✗	✗	✗	✗	✗	✗	✗	✗	✓	✓	–	✗	✗	–	15
Macharla Srivyshnav	64	224	82	106.9	✗	✗	–	✗	–	–	✗	✗	✓	✗	✗	–	✗	–	✗	✗	✗	16
Students who lost it					6	6	7	9	5	5	10	6	6	10	10	6	5	6	6	10	10	

6 Subject scores of the top 10

Estimated subject scores (≈) from right / wrong / skipped answers; partial credit on multiple-correct questions is not in the source data, so these can run a few marks above the official total. The weakest subject (lowest % of its maximum) is highlighted. **Subject maximum** leaves out the deleted / ungraded questions: Maths 50 = 60 minus Q13 (4), Q14 (3), Q17 (3); Physics 56 = 60 minus Q29 (4); Chemistry 57 = 60 minus Q50 (3).

STUDENT	PROG. RANK	AIR	TOTAL	AVG /180	Maths score · right / wrong / skipped	Physics score · right / wrong / skipped	Chemistry score · right / wrong / skipped	WEAKEST
Ruperao Shritan	14	38	117	122.5	≈ 44/50 12 / 2 / 0	≈ 32/56 10 / 6 / 0	≈ 47/57 13 / 2 / 1	Physics
Kolukula H. S. N. Hariharan	17	44	116	132.7	≈ 34/50 11 / 3 / 0	≈ 38/56 11 / 4 / 1	≈ 44/57 12 / 4 / 0	Physics
Kavuru Sahithi	27	89	104	127.2	≈ 43/50 12 / 2 / 0	≈ 32/56 9 / 7 / 0	≈ 38/57 11 / 5 / 0	Physics
Fareeha Shaik	29	97	102	90.8*	≈ 37/50 11 / 3 / 0	≈ 28/56 8 / 8 / 0	≈ 43/57 12 / 4 / 0	Physics
Karthikeya Silver	38	114	95	109.5	≈ 40/50 11 / 3 / 0	≈ 28/56 8 / 8 / 0	≈ 28/57 9 / 6 / 1	Chemistry
Sriyan Brahmakanti	52	214	86	93.1	≈ 36/50 10 / 4 / 0	≈ 19/56 7 / 9 / 0	≈ 37/57 11 / 4 / 1	Physics
Chitluri Nayan Swaroop	54	200	85	68.9	≈ 34/50 10 / 4 / 0	≈ 22/56 7 / 7 / 2	≈ 31/57 10 / 6 / 0	Physics
G Umesh Babu	58	242	84	99.3	≈ 36/50 10 / 4 / 0	≈ 22/56 8 / 8 / 0	≈ 33/57 10 / 5 / 1	Physics
Nakka Mahidhar	59	209	84	98.9	≈ 32/50 9 / 4 / 1	≈ 22/56 8 / 8 / 0	≈ 32/57 10 / 4 / 2	Physics
Macharla Srivyshnav	64	224	82	106.9	≈ 36/50 10 / 3 / 1	≈ 15/56 4 / 6 / 6	≈ 34/57 10 / 5 / 1	Physics

Average. "Avg" = the student's average over the 6 weekly Advanced tests (WTA) before this Part Test, out of 180 (those tests are marked out of 240; averages are scaled to 180 to match this paper – × 180/240, percentages unchanged), from the campus Mains & Advance averages sheet. * Sat fewer weekly tests (tests sat / conducted): Fareeha Shaik (5/6).
Ranks. Program rank = rank within Pragya G8 (EXI) Achievers across all campuses (from the final marks sheet; the top 10 are picked by this rank). **AIR** = All-India rank across the whole Class 8 programme (all batches, all campuses), from the institute's All India Marks Analysis sheet.
How to read this report. "Lost" means the student did not get the question right – either answered it wrong or left it blank. A question is **high-impact** when 5 or more of the top 10 lost it. **"Marks lost"** = marks the top 10 together gave away: the question's marks plus its negative mark for each wrong answer (single correct 3 + 1 = 4, multiple correct 4 + 2 = 6, integer 4, paragraph 3), and the question's marks for each skipped one. "Per student" divides that by 10.
Final key applied. Deleted in the final key: Q14, Q17, Q50. Not deleted in the final key, but the final marks sheet has no right / wrong / skipped entry for any student, so they cannot be counted: Q13 (key 4046), Q29 (key 9). Answer revised in the final key and counted as marked in the final marks sheet: Q25 (key 1 OR 2), Q48 (key 2 OR 5), Q51 (key 3). Left-out questions are marked "del" or "n/a" in the charts. Student totals and program ranks are as printed in the final marks sheet.
Chapter tags were assigned by reading each question; please correct any that your department classifies differently.
Note on Advanced marks. The paper uses partial marking for multiple-correct questions, but the source data only says right / wrong / skipped. Marks-lost figures use the paper's marking scheme (shown on every question) and are approximate; the counts of students who lost each question are exact.

7 Maths – high-impact questions

4 questions, worst first
cropped from the question paper

Q16 Logarithms – Log equations – A + B + C

Maths · Paragraph · marking +3 / 0 · marks lost by the top 10: 9 wrong × 3 = 27

Lost by 9/10
9 wrong · 0 skipped

PARAGRAPH-II

Let **A** denote the product xyz where x, y, z are solutions of the system of equations

$\log_3 x = \log_3 5 - \log_3 7, \log_5 y = \log_5 7 - \log_5 3, \log_7 z = \log_7 3 - \log_7 5$

Let **B** denote the sum of square of the solutions of the equation

$\log_2\left(\log_2 x^6 - 3\right) - \log_2\left(\log_2 x^4 - 5\right) = \log_2 3$

Let **C** denote the characteristic of the number K, where

$K = \log_2\left(\log_2 3\right) - \log_2\left(\log_4 3\right) + \log_2\left(\log_4 5\right) - \log_2\left(\log_6 5\right) + \log_2\left(\log_6 7\right) - \log_2\left(\log_8 7\right)$

16. The value of A+B+C is____?

Wrong: Ruperao Shritan, Kavuru Sahithi, Fareeha Shaik, Karthikeya Silver, Sriyan Brahmakanti, Chitluri Nayan Swaroop, G Umesh Babu, Nakka Mahidhar, Macharla Srivyshnav

Skipped: –

Q15 Sequences & Series – AP: ratio of sums of 14 terms

Maths · Paragraph · marking +3 / 0 · marks lost by the top 10: 5 wrong × 3 + 2 skipped × 3 = 21

Lost by 7/10
5 wrong · 2 skipped

Sum of n terms of two different Arithmetic progressions are in the ratio 5n+4:9n+6 then

15. The ratio of the sum of their first 14 terms is Q, then [100Q] is _____(where [.] is G.I.F)

Wrong: Ruperao Shritan, Karthikeya Silver, Sriyan Brahmakanti, Chitluri Nayan Swaroop, G Umesh Babu

Skipped: Nakka Mahidhar, Macharla Srivyshnav

Q4 Logarithms – Logs in AP – solve for x

Maths · Single correct · marking +3 / −1 · marks lost by the top 10: 6 wrong × 4 = 24

Lost by 6/10
6 wrong · 0 skipped

4. If $\log_3 2, \log_3\left(2^x - 5\right)$ & $\log_3\left(2^x - \frac{7}{2}\right)$ are in AP then the value of x is?

A)2 B)3 C)4 D)2&3

Wrong: Kolukula Hema Shiva Naga Hariharan, Karthikeya Silver, Sriyan Brahmakanti, G Umesh Babu, Nakka Mahidhar, Macharla Srivyshnav

Skipped: –

Q10 Quadratic Equations – Roots of two linked quadratics

Maths · Integer · marking +4 / 0 · marks lost by the top 10: 6 wrong × 4 = 24

Lost by 6/10
6 wrong · 0 skipped

10. Let a and b be roots of the equation $x^2 - 10cx - 11d = 0$ and the roots of $x^2 - 10ax - 11b = 0$ are c and d; then the value of $\frac{a+b+c+d}{605}$ is____?(Given $a \neq b \neq c \neq d \neq 0$)

Wrong: Kavuru Sahithi, Sriyan Brahmakanti, Chitluri Nayan Swaroop, G Umesh Babu, Nakka Mahidhar, Macharla Srivyshnav

Skipped: –

Physics – high-impact questions

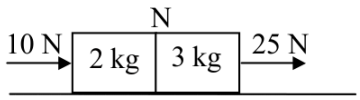
9 questions, worst first
cropped from the question paper

Q27 Laws of Motion – Normal reaction between pushed blocks

Physics · Integer · marking +4 / 0 · marks lost by the top 10: 10 wrong × 4 = 40

Lost by 10/10
10 wrong · 0 skipped

27. The normal reaction between 2 kg and 3 kg block is



Wrong: Ruperao Shritan, Kolukula Hema Shiva Naga Hariharan, Kavuru Sahithi, Fareeha Shaik, Karthikeya Silver, Sriyan Brahmakanti, Chitluri Nayan Swaroop, G Umesh Babu, Nakka Mahidhar, Macharla Srivyshnav

Skipped: –

Q31 Vectors – Carrom passage – displacement centre → edge

Physics · Paragraph · marking +3 / 0 · marks lost by the top 10: 10 wrong × 3 = 30

Lost by 10/10
10 wrong · 0 skipped

PASSAGE-1:

A Corrom board $(4\,ft \times 4\,ft\ square)$ has queen at the centre. The queen , hit by the striker moves to the front edge, rebounds and goes in the hole behind the striking line.
Answer the following question.

31. The magnitude of the displacement of the queen from the centre to the front edge is $\frac{2}{3}\sqrt{a}\,ft$, the value of a is _____

Wrong: Ruperao Shritan, Kolukula Hema Shiva Naga Hariharan, Kavuru Sahithi, Fareeha Shaik, Karthikeya Silver, Sriyan Brahmakanti, Chitluri Nayan Swaroop, G Umesh Babu, Nakka Mahidhar, Macharla Srivyshnav

Skipped: –

Q32 Vectors – Carrom passage – displacement edge → hole

Physics · Paragraph · marking +3 / 0 · marks lost by the top 10: 10 wrong × 3 = 30

Lost by 10/10
10 wrong · 0 skipped

PASSAGE-1:

A Corrom board $(4\,ft \times 4\,ft\ square)$ has queen at the centre. The queen , hit by the striker moves to the front edge, rebounds and goes in the hole behind the striking line.
Answer the following question.

32. The magnitude of the displacement of the queen from the front edge to the hole is $\frac{4}{3}\sqrt{b}\,ft$, the value of b is_____

Wrong: Ruperao Shritan, Kolukula Hema Shiva Naga Hariharan, Kavuru Sahithi, Fareeha Shaik, Karthikeya Silver, Sriyan Brahmakanti, Chitluri Nayan Swaroop, G Umesh Babu, Nakka Mahidhar, Macharla Srivyshnav

Skipped: –

Q30 Kinematics – a from v(t) at a given position

Physics · Integer · marking +4 / 0 · marks lost by the top 10: 6 wrong × 4 = 24

Lost by 6/10
6 wrong · 0 skipped

30. A body starts from origin and moves along x axis such that at any instant velocity is $v_t=4t^3-2t$ where t is in second and v_t is in ms^{-1}. The acceleration of the particle when it is 2m from the origin is $(20+K)\,ms^{-2}$, where K is

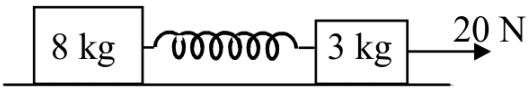
Wrong: Kolukula Hema Shiva Naga Hariharan, Fareeha Shaik, Sriyan Brahmakanti, Chitluri Nayan Swaroop, G Umesh Babu, Nakka Mahidhar

Skipped: –

Q28 **Laws of Motion** – Spring-connected blocks – acceleration
Physics · Integer · marking +4 / 0 · marks lost by the top 10: 5 wrong × 4 + 1 skipped × 4 = 24

Lost by 6/10
5 wrong 1 skipped

28. If the acceleration of 3 kg block is 4 m/s^2 to the left. What is the magnitude of acceleration of 8 kg block.



Wrong: Ruperao Shritan, Karthikeya Silver, Chitluri Nayan Swaroop, Nakka Mahidhar, Macharla Srivyshnav **Skipped:** Kolukula Hema Shiva Naga Hariharan

Q33 **Kinematics** – Train passage – retardation to avoid collision
Physics · Paragraph · marking +3 / 0 · marks lost by the top 10: 4 wrong × 3 + 2 skipped × 3 = 18

Lost by 6/10
4 wrong 2 skipped

PASSAGE-II:

The driver of a train A moving with a uniform speed of 144 km/h sights another train B, 1km ahead of him. The train B is moving with a uniform speed of 108 km/h. the driver of the train A immediately applies brakes producing a constant retardation and just manages to avoid a collision.

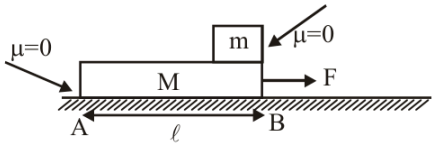
33. What is the retardation of the train A(in ms^{-2})_____

Wrong: Kavuru Sahithi, Fareeha Shaik, Karthikeya Silver, Sriyan Brahmakanti **Skipped:** Chitluri Nayan Swaroop, Macharla Srivyshnav

Q23 **Laws of Motion** – Block on block, no friction – motion of m
Physics · Multiple correct · marking +4 / −2 · marks lost by the top 10: 4 wrong × 6 + 1 skipped × 4 = 28

Lost by 5/10
4 wrong 1 skipped

23. In the figure small block is kept on M, then



A) the acceleration of m w.r.t. ground is $\frac{F}{m}$

B) the acceleration of m w.r.t. ground is zero

C) the time taken by m to separate from M is $\sqrt{\frac{2\ell m}{F}}$

D) the time taken by m to separate from M is $\sqrt{\frac{2\ell M}{F}}$

Wrong: Ruperao Shritan, Sriyan Brahmakanti, G Umesh Babu, Nakka Mahidhar **Skipped:** Macharla Srivyshnav

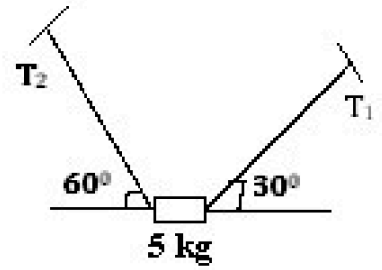
Q24 **Laws of Motion** – Tensions in two inclined strings

Physics · Multiple correct · marking +4 / −2 · marks lost by the top 10: 4 wrong × 6 + 1 skipped × 4 = 28

Lost by 5/10

4 wrong 1 skipped

24. A body of mass 5 kg is suspended by the strings making angles 60^0 and 30^0 with the horizontal as shown in the figure($g = 10ms^{-2}$). Then



A) $T_1 = 25N$ B)$T_2 = 25N$ C) $T_1 = 25\sqrt{3}N$ D) $T_2 = 25\sqrt{3}N$

Wrong: Sriyan Brahmakanti, Chitluri Nayan Swaroop, G Umesh Babu, Nakka Mahidhar **Skipped:** Macharla Srivyshnav

Q34 **Kinematics** – Train passage – time of retardation

Physics · Paragraph · marking +3 / 0 · marks lost by the top 10: 4 wrong × 3 + 1 skipped × 3 = 15

Lost by 5/10

4 wrong 1 skipped

PASSAGE-II:

The driver of a train A moving with a uniform speed of 144 km/h sights another train B, 1km ahead of him. The train B is moving with a uniform speed of 108 km/h. the driver of the train A immediately applies brakes producing a constant retardation and just manages to avoid a collision.

34. For how long is this retardation produced(in sec)_____

Wrong: Kavuru Sahithi, Fareeha Shaik, Karthikeya Silver, Macharla Srivyshnav **Skipped:** Chitluri Nayan Swaroop

Chemistry – high-impact questions

4 questions, worst first
cropped from the question paper

Q48 Mole Concept & Basics – Empirical formula from % composition

Chemistry · Paragraph · marking +3 / 0 · marks lost by the top 10: 10 wrong × 3 = 30

Lost by 10/10
10 wrong 0 skipped

PARAGRAPH-I

An unknown copper oxide is analysed. A 12.0 g sample contains 9.6 g Cu, so the remaining 2.4 g is oxygen. Taking Cu = 64 and O = 16, the moles of Cu and O are equal

48. What is the sum of the simplest whole-number ratio of Cu:O?

Wrong: Ruperao Shritan, Kolukula Hema Shiva Naga Hariharan, Kavuru Sahithi, Fareeha Shaik, Karthikeya Silver, Sriyan Brahmakanti, Chitluri Nayan Swaroop, G Umesh Babu, Nakka Mahidhar, Macharla Srivyshnav

Skipped: –

Q51 Atomic Structure – Hydrogen-like ion – find Z from photon energies

Chemistry · Paragraph · marking +3 / 0 · marks lost by the top 10: 6 wrong × 3 + 4 skipped × 3 = 30

Lost by 10/10
6 wrong 4 skipped

PARAGRAPH-II

When electron jumps from lower orbit to higher orbit energy is gained and when electron jumps from higher orbit to lower orbit energy is released.

51. A hydrogen like species (atomic number Z) is present in a higher excited state of quantum number n. This excited atom can make a transition to the first excited state by successive emission of two photons of energies 10.20 eV and 17.0 eV respectively. Alternatively the atom from the same excited state can make a transition to the second excited state by successive emission of two photons of energy 4.25 eV and 5.95 eV respectively. Determine the value of Z.

Wrong: Kolukula Hema Shiva Naga Hariharan, Kavuru Sahithi, Fareeha Shaik, Karthikeya Silver, Chitluri Nayan Swaroop, Macharla Srivyshnav

Skipped: Ruperao Shritan, Sriyan Brahmakanti, G Umesh Babu, Nakka Mahidhar

Q39 Mole Concept & Basics – Meaning of 1 gram-atom of nitrogen

Chemistry · Multiple correct · marking +4 / −2 · marks lost by the top 10: 6 wrong × 6 = 36

Lost by 6/10
6 wrong 0 skipped

39. 1 g atom of nitrogen represents:

(A) 6.02×10^{23} N_2 molecules
(B) 22.4 litre of N_2 at N.T.P.
(C) 11.2 litre of N_2 at N.T.P.
(D) 14 g of nitrogen

Wrong: Karthikeya Silver, Sriyan Brahmakanti, Chitluri Nayan Swaroop, G Umesh Babu, Nakka Mahidhar, Macharla Srivyshnav

Skipped: –

Q35 Mole Concept & Basics – Limiting reagent over two-step reaction

Chemistry · Single correct · marking +3 / −1 · marks lost by the top 10: 3 wrong × 4 + 3 skipped × 3 = 21

Lost by 6/10
3 wrong 3 skipped

35. 5 moles of A, 6 moles of Z and mixed with sufficient amount of C to produce final product F. How many maximum moles of 'F' can be produced as per the given sequence of reaction ?

$A + 2Z \rightarrow B$; $B + C \rightarrow Z + F$

(A) 3 (B) 2 (C) 4 (D) 5

Wrong: Ruperao Shritan, Sriyan Brahmakanti, G Umesh Babu

Skipped: Karthikeya Silver, Nakka Mahidhar, Macharla Srivyshnav